Lab program-29:

29. Write a C program to simulate the solution of Classical Process Synchronization Problem.

Code:

```c
#include <stdio.h>
#include <stdlib.h>

int mutex = 1;
int full = 0;
int empty = 10;
int x = 0;

void producer(void) {
    --mutex;
    ++full;
    --empty;
    ++x;
    printf("\nProducer produces item %d", x);
    ++mutex;
}

void consumer(void) {
    --mutex;
    --full;
    ++empty;
```

```c
    printf("\nConsumer consumes item %d", x);
    --x;
    ++mutex;
}

int main(void) {
    int choice;

    while (1) {
        printf("\n\n1. Press 1 for Producer");
        printf("\n2. Press 2 for Consumer");
        printf("\n3. Press 3 for Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                if (mutex == 1 && empty != 0)
                    producer();
                else
                    printf("\nBuffer is full!");
```

```
                break;
            case 2:
                if (mutex == 1 && full != 0)
                    consumer();
                else
                    printf("\nBuffer is empty!");
                break;
            case 3:
                return 0;
            default:
                printf("\nInvalid choice!");
        }
    }
}
```

OUTPUT:

```
1. Press 1 for Producer
2. Press 2 for Consumer
3. Press 3 for Exit
Enter your choice:1

Producer producesitem 1
Enter your choice:1

Producer producesitem 2
Enter your choice:1

Producer producesitem 3
Enter your choice:2

Consumer consumes item 3
Enter your choice:3

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Process exited after 6.797 seconds with return value 0
Press any key to continue . . .
```